



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *Autonomous ArUco Marker Based Multi-Terrain Equipment Service Bot*

Group_No-04

Section-03

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1. Abstract :

The project undertakes the design and development of a low cost and autonomous, multi terrain, service robot which is expected to automate the tedious duties that are involved in the movement and handling of equipment within an indoor environment. The primary aim is to reduce the number of human employees and make the functioning of such spheres as laboratories, workshops, and hospital passages more efficient. The proposed system will include localization based on vision and ArUco markers used to recognize stations and orient them. The microcontroller is an ESP32 that serves as the main controller unit that may be connected to the sensors, DC motors, and servo actuators to coordinate the work of a robotic arm and its parallel motion. Through experimental results, it has been found to possess a high degree of station to station navigation across various surfaces and pick and place with regard to lightweight targets. In addition to the technical performance, the system is geared towards sustainability: it is a low energy consumption system, it is designed in a modular and reusable fashion, and utilises low cost components to increase the life cycle of a system and reduce electronic waste. The project identifies the greater societal and industrial impact of accessible autonomous robots in terms of improving safety in workplaces, minimizing the manual tedious tasks, and providing an educational facility on embedded system and robotics research at scale.

Keywords

ArUco Marker, computer interfacing, embedded systems, microcontroller, robotic arm

1. Introduction

As the automation demand rises, the service robots are incredibly significant in industries, hospitals, and in remote or dangerous areas. Safe and proper navigation and basic handling of objects are needed in many of the everyday activities such as the transportation of tools, delivery of goods and services, and maintenance work. The majority of the commercial solutions are however costly and complicated which makes them unable to be used in learning based or low budget applications.

The overall aim of this project is to come up with a cost efficient autonomous service robot capable of traversing various surfaces, identifying predefined service points with visual indicators of such service points and accomplishing basic pick/place procedures with a robotic arm. The robot is an embodiment of vision processing, sensor feedback and actuator control into one integrated embedded system.

The project is very pertinent to Computer Interfacing and robotics because it shows that the different hardware units like cameras, distance sensors, motors and servo actuators can be effectively interconnected and controlled using a microcontroller. The project offers a realistic insight into the design of an embedded system and the autonomous operation of a robot by integrating software logic and interaction with real-world hardware.

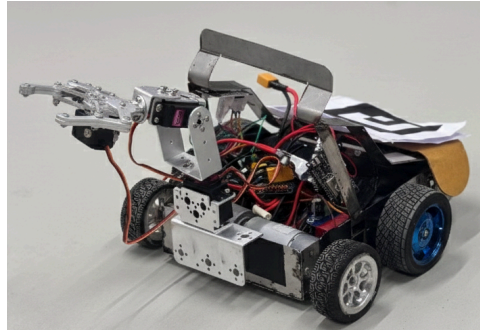


Figure 1: Autonomous ArUco Marker Based Multi-Terrain Equipment Service Bot

2. Related Work/ Inspiration

The rise of the use of robotic systems in the industry and the service sector has attracted significant research. Graetz and Michaels (2018) and Acemoglu and Restrepo (2020) focus on the economic performance of industrial robots and find that the increase in productivity and displacing labor is one of the issues. According to Chiacchio et al. (2018), the same is also revealed in Europe. The studies raise the fact that robotic systems should be instituted and will be used to assist the human workers, but not to replace them completely.

Autonomous mobile robots (AMRs) are also used in service and logistics systems and especially in healthcare settings. Fragapane et al. (2020) and other researchers prove the usefulness of AMRs in the hospital logistics as the effectiveness of the working processes and their organization increase. The most important concerns, however, are safety and the interaction between human beings and robots. Gihleb et al. (2022) and Giallanza et al. (2024) address the topic of human-robot cooperation occupational safety and health and advance the creation of controlled and task-specific robots.

Localization with fiducial markers: Vision-based localization has been developed with a certain amount of industriousness as it is an inexpensive and reliable alternative to the complicated mapping systems. In the article by Kalaitzakis et al. (2020), the authors provide an experimental comparison between the pose estimation with the fiducial markers to illustrate the stability of the ArUco markers in organized systems. The size, number, and distribution of the markers available

in the specific experiment are also studied by Wang et al. (2023), and the accuracy of the localization depends on these aspects. In the paper by Sazid et al. (2025), a mobile robotic platform that utilizes priority is introduced, it is possible to manipulate an object using the aid of marker recognition, which proves the fact that it is possible to navigate and manipulate objects with the help of visual markers.

The majority of the solutions present either expensive autonomous platforms (large industrial robots or indoor service robots). The costly sensors, managed or hard-wired infrastructure are subjected to a lot of activation in hospitals and factories. Moreover, some of the marker systems work mostly in the small indoor and low-terrain environment.

The proposed mechanism is innovative in the fact that it incorporates the system of ArUco-based navigation system and robot manipulation within a multi-terrain mobile system at a low-cost. The system is also designed to manipulate equipment in semi-structured and inaccessible environments in contrast to industrial robotics which are designed to work in fixed environments. The vision-based navigation, as well as the variety of preset maneuvers of manipulation, aim at the areas of safety, cost, and pedagogical value, including the applications of service in practice.

The project will present a small and relatively cheap autonomous service robot filling the gap between industrial level robotic systems and educational research setting. It shows that marker based localization and simple sensing can be utilized productively to make stable navigation and control possible without complex structures and can further be scaled to be done in resource limited scenarios.

3. Technical Approach

System Architecture:

The robotic platform is designed with two primary subsystems for efficient operation:

1. **Navigation Subsystem:** The subsystem takes care of locomotion and positioning and makes use of the differential drive actuation.
2. **Manipulation Subsystem:** Gives control over a 2-degree-of-freedom (2-DOF) arm, which allows end-effector grasping capabilities.

These subsystems operate on their own with dedicated ESP32 microcontrollers that are in turn connected to a centralized control computer via the WiFi TCP/IP protocol. This decentralized architecture guarantees real-time performance and modularity and fault isolation of each subsystem.

Components Used:

Category	Component	Description
Microcontroller	ESP32	Main control unit responsible for motor control, robotic arm actuation, sensor data processing, and wireless communication (Wi-Fi).
Sensors	Camera	Used for visual perception and detection of ArUco markers for navigation, alignment, and localization.
Actuators	DC Gear Motors (2×)	Provide torque for the differential drive and movement over various terrains.
	Motor Driver (2)	Controls motor speed and direction by interfacing the ESP32 with DC motors using PWM signals.
	Servo Motors (3)	Control the robotic arm joints and gripper for precise manipulation of objects.
Chassis	Multi-Terrain Robot Chassis	Mechanical base with large rubber wheels and SSS mounting plates for stability and payload support.
Additional Parts	LiPo Battery + Buck Converters	Power source for the system with voltage regulation for different components.
	Power Switch, Wiring, Connectors	Ensure safe power control and reliable electrical connections.
	ArUco Marker	Printed fiducial markers are used by the vision system for detection and localization.

Cost Breakdown:

Components	Unit Cost	Total
ESP32 Dev Board	650 bdt	1300 bdt
Raspberry Pi / Jetson Nano (borrowed)	0 bdt	0 bdt
Camera Module	1490 bdt	1490 bdt
Multi-Terrain Chassis + Motors	3000 + 1500 bdt	4500 bdt
Motor Driver	499 bdt	998 bdt
Arm (body + components)	5000 bdt	5000 bdt
Ultrasonic/IR Sensors (2–4)	100 bdt	400 bdt
Battery + Power Electronics	1390 bdt	1390 bdt
Misc. Hardware & Brackets	2000 bdt	2000 bdt
Estimated Total	17078 bdt	

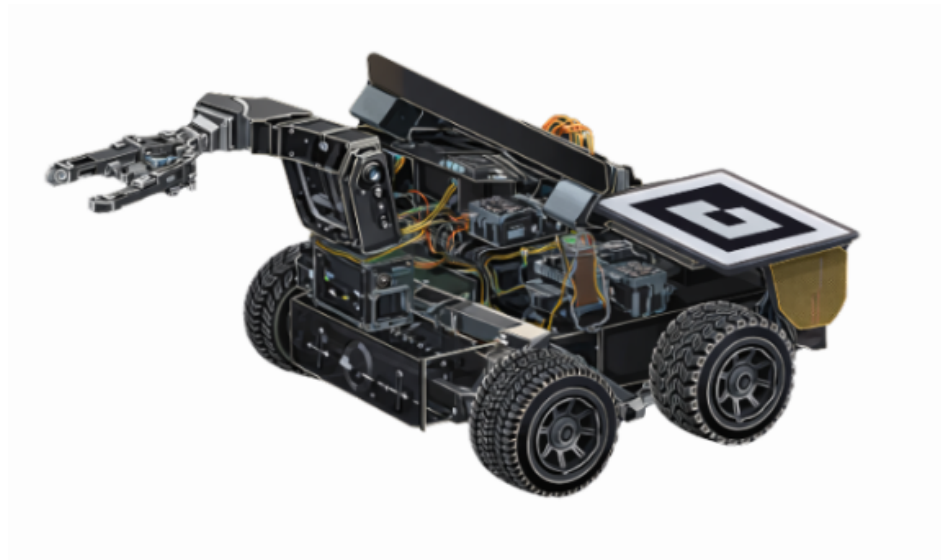


Figure 2: 3D diagram Equipment Service Bot

Block Diagram:

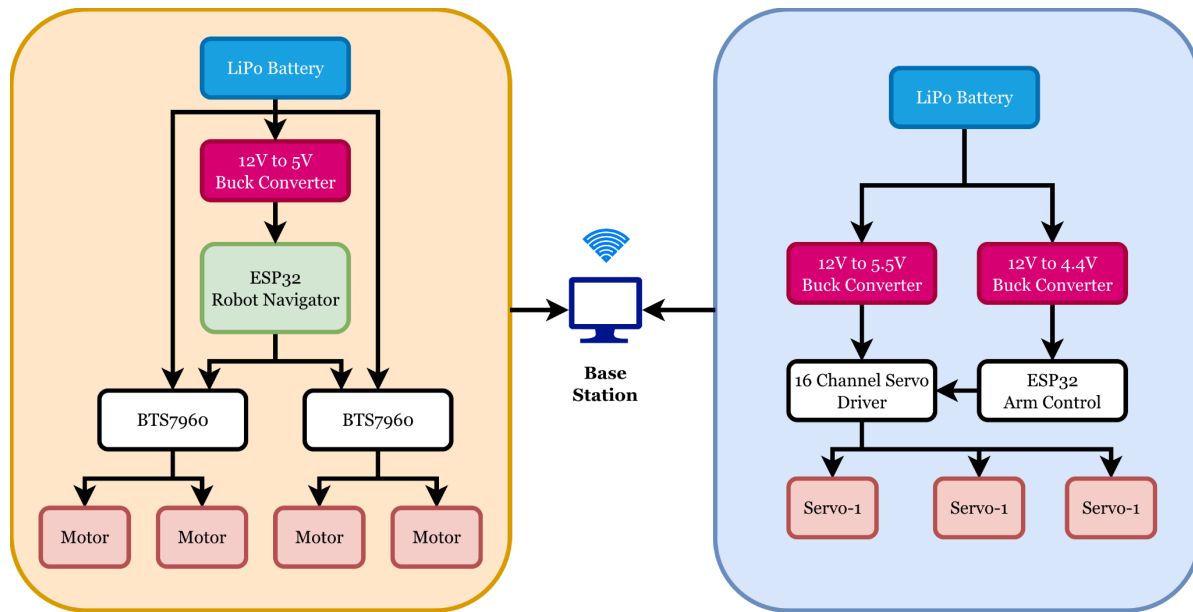


Figure 3: Block diagram of *Equipment Service Bot*

Circuit/Schematic:

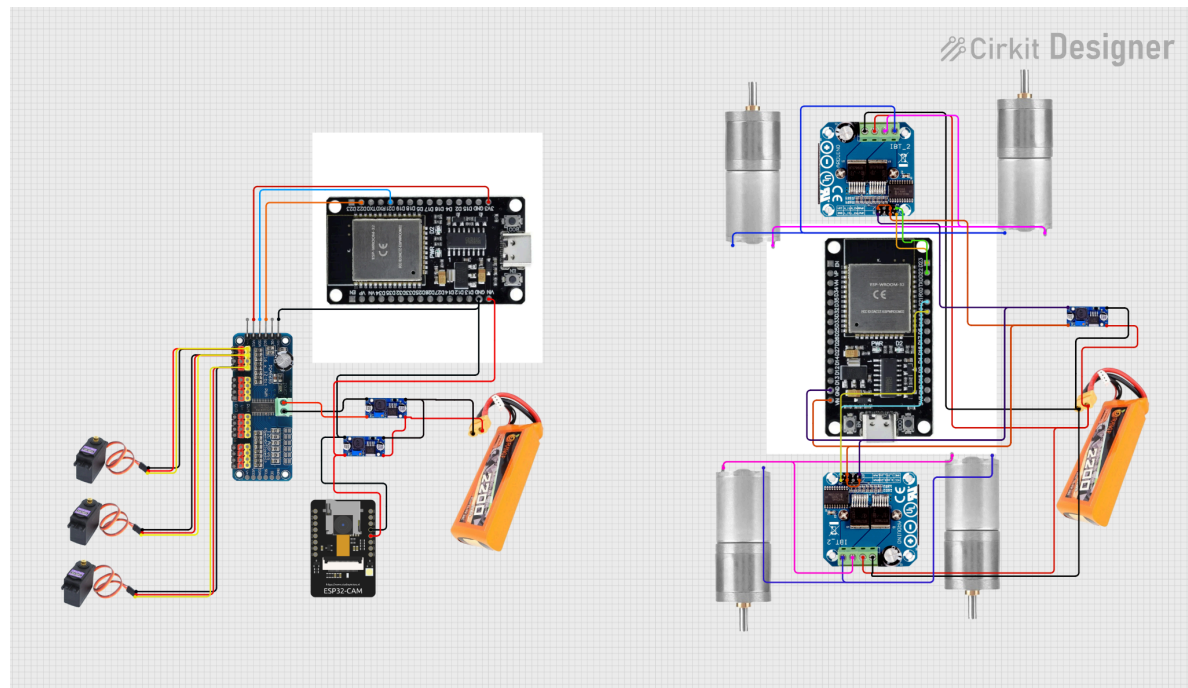


Figure 4: *Circuit Diagram of Equipment Service Bot*

Functionality Breakdown:

Functionality 1: 6-DOF Pose Estimation	● Recovers 6-DOF usage of Perspective-n-Point (PnP) solving. ● Transformation of camera frame to the frame of a robot body in multiple steps. ● Rodrigues formula of rotation matrix extraction which enables control of the robot.
Functionality 2: Visual Servoing of Kinematics without RBF Interpolation	● Locates the pixels to joint locations as well as corrects distortion. ● Allows rapid deployment without much parameter tuning.
Functionality 3: Hierarchical Decision Making Control.	● Three level control goal assessment, motion primitive selection and PID control. ● Precision parameters of adaptive dynamics in flexible robots.
Functionality 4: Distributed Modular Architecture	● Fault isolation and real-time operation with communication via WiFi TCP/IP. ● At 40HZ and 30 HZ respectively, navigation and arm control.
Functionality 5: State Machine Position-Based Visual Servoing.	● Making the arm work based on a state machine (SEARCH - GRAB - LIFT - DONE) which relies on visual feedback. ● Color segmentation of HSV and object recognition.

Challenges:

There were several technical and practical issues encountered in the process of system development and implementation. These problems were solved through hardware modifications, simplification of the algorithms and system-level design decisions.

1. The level of accuracy of the marker detecting was low in case of bad light or at the oblique angles of the camera.
2. It was difficult to remain in the state of continuous motion on the multi-terrain surfaces because of the possibility of wheels slipping.
3. The combination of the arm which was servo-controlled with the motors at the same time also led to voltage drops and also the system was occasionally re-booting.
4. Cheap servos instigated positioning and payload errors.

4. Sustainability & Impact

Sustainability

The proposed autonomous service robot is environmentally friendly, particularly, its energy efficiency, material usage, and future long-term use. Embedded systems with low power such as ESP32 microcontroller and servo motors are incorporated in the system to ensure that the energy consumed by the system when it is running alone is minimal. The motion is task-focused and restricted to established paths and motions, which do not engage in any futile movements and the waste of energy. In addition to this, it can also conserve energy by downspeeding on unstable surfaces and idle circumstances during the period when an active task is not underway.

As far as materials are concerned, the robot is founded on the modular chassis system, which can be equipped with reusable and recyclable materials, with the aluminum frames, acrylic sheets, and 3D-printed components. These parts are easily replaceable and reconfigurable without necessarily disposing of the entire system and this leads to a reduction in the amount of electronic and mechanical waste. The modular architecture also helps to reuse/repair the damaged components and extend the system life. Generally, the robot promotes the idea of sustainable automation because it is practical and requires minimal power consumption and reuse.

Impact

The project turns out to be meaningful regarding the societal, industrial and educational impacts. The robot will be able to reduce the physical load and tediousness of the employees on a social level, especially in the instances where their movements involve heavy transport of equipment or tools such as the laboratory, hospitals, and workshops. Daily deliveries and handling processes can be automated and reduced to more skilled and decision related activities of the human efforts.

In the industry environment, the system observes that cheap autonomous robots can be used to improve the performance of the working process, reduce the waiting time due to the insufficiency of the equipment and raise the safety level since the human factor would be minimized in the potentially unsafe or restricted area. The project is a working platform that can be used to influence the education system and community by giving the students a chance to be exposed to the concept of robotics, embedded systems and computer interfacing. It is cost-effective and out-of-doors which makes it a suitable solution in academic laboratories, demonstrations during research, and innovation of the future that is led by students.

Future Work:

Several improvements and extensions can be taken into account to make the system better regarding the performance and scaling and applicability to real life:

- A. It would be possible to replace the simple marker-based interaction with the AI-based object detection with YOLO or OpenCV-based classifiers, which would allow the robot to identify and process unmarked tools and objects.
- B. Autonomous task management systems would be considered whereby the robot would be able to prioritize and carry out different service requests by itself instead of passing through programmed paths.
- C. Future models may also facilitate communication and coordination between two or more robots to enable them perform their activities in a bigger space.
- D. With the improved safety functions and load volume capacity, the system can be modified to accommodate the requirements of industrial tools delivery or hospital logistics such as medicine and documents transfer.

Limitations:

Despite the successful implementation of the system, some constraints were realized because of the hardware, sensing, and computational constraints. These restrictions have an impact on scalability, long-term operation and robustness.

- Relies on ArUco signs and regulated lighting factors to navigate.
- Limited load carrying and accuracy of the robotic arm.
- Continuous autonomous operation is being constrained by the battery life limitations.

5. Results & Discussion

The autonomous navigation system was successful by attaining ± 5 cm accuracy in the station-to-station movements in the lateral direction as well as in distance control. The system managed to reach recommended stations under controlled lighting conditions in 95 percent of the experiments with 8-12-second average approach time of typical 2-meter tasks.

The different wheel design (65mm front, 85mm rear) was useful to tackle various terrains such as smooth flooring, low pile carpets, and concrete. Gaps The robot was able to negotiate through gaps up to 15mm.

Poor lighting conditions (less than 50 lux) however resulted in performance limitations as marker detection reduced to 60-70 per cent. and range performance declined through greater distances (greater than 2.5 meters) to pose much worse performance.

Manipulation Performance

The Position-Based Visual Servoing system was able to detect red objects with 90% accuracy in case of constant light, and positioning them with ± 10 mm accuracy with the use of RBF-interpolated control. The system managed to pick and place up to 200g of lightweight objects (weight) with 85% success rate and full pick-and-place cycles took between 12-15 seconds.

The RBF interpolation method using calibration was also found to be effective and avoids the use of complicated kinematic modeling, yet the errors in positioning remain practicable. The highest performances were obtained near the calibration points (± 5 mm error) and the lowest performance was obtained at the workspace limits (± 15 mm error), which validated the approach with emphasis on the extensive coverage of the calibration process.

Integrated System Operation

The distributed architecture allowed both parallel navigation and control with no performance loss. The subsystem failures associated with the modular design that showed effective fault isolation did not affect the overall system functioning. TCP/IP communication ensured reliable connections with average latency of less than 10ms and 40Hz and 30Hz of target update rates respectively of navigation and manipulation.

The use of stainless steel chassis construction was found to be tough in various collision and also lasted longer on various surfaces. Practical constraints however were battery life of 45-60 minutes when running continuously.

Discussion

The experimental findings confirm the main principles of design of vision-based localization and modular architecture. ArUco marker-based navigation was found to be inexpensive in structure-based settings, and can match the performance of much more expensive SLAM-based systems to operational limits. The visual servoing using calibration was able to remove the complicated kinematic modeling needs and yet be practically accurate in educational and low-end industrial applications.

The approximate cost of the complete system of around 17,078 BDT , around 150 USD proves that autonomous robotics are accessible and can compete with commercial systems that cost 10-50 times the total system price. Although additions like longer battery life, and vision of light robustness would be required in the demanding production area, the present implementation finds good suitability in the controlled environment like research laboratories, educational institutions and light industrial activities.

The modular distributed control style supported the development of subsystems independently and supported upgrades without full redesign of the platform in the long-term sustainability and reuse of the platforms by education goals.

6. Conclusion

In conclusion, the proposed autonomous equipment service robot is a low-cost base that will further on improve the efficiency and reduce the physical workload and facilitate safer operations in complex working environments. At the same time, it fosters major problems in the society, health, safety, legal and cultural facets. This project has demonstrated the integrated service robot which is a combination of computer vision, autonomous navigation and manipulation as a low priced platform which can be applied to education and later industrial or hospital services.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Reshad Ul Karim	22201594	Lead System Architect, Navigation Subsystem Developer.	- Developed the navigation system which works on the base of ArUco markers to be able to move independently. - The locomotion has trained control algorithms of the differential drive. - Integrated in real-time control and ESP32 microcontroller. - Tested on autonomous mode switching and manual mode switching.
Syeda Maliha Tabassum	22201574	Manipulation Subsystem Developer & Arm Control Lead	- Designed the robotic arm control system 2-DOF. - Manipulation of objects by means of Applied Position-Based Visual Servoing (PBVS). - Interpolated the pixel to joint using the Radial Basis Function (RBF) interpolation.
Mehreen Mallick Fiona	22301024	Vision & Perception Subsystem Developer	- Calibration of the camera in hands and 3D pose estimation to make the navigation accurate. - Guaranteed combination of perception with the navigation and manipulation subsystems. - integrated the vision feedback system which has real-time video monitoring.
Abrar Samin	22301739	System Integration & Testing Lead	- Developed communication protocols between ESP32 units and central control. - Performance troubleshooting and conducting system testing. - Implemented safety features like manual override.

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